BDA400 / DATA SCIENCE TOOLS AND TECHNIQUES

Technical Analysis Using R

Development Stage | Assignment 5

Prepared by

Sunny Khurana

Student ID: **664425271**

Project brief

A core-R implementation of nine technical-analysis indicators, backed by deterministic tests, a reproducible two-symbol OHLCV fallback, time-frame and data-source filtering, indicator-based stock selection, and a simple moving-average trading rule.

Repository location

https://github.com/sunny-khurana-26/TechnicalAnalysis/tree/main/Assignment%205%20-%20Technical%20Analysis%20Using%20R%20Development%20Stage

Public repository path for this assignment

Deliverable package

Nine separate R scripts | Base-R test runner | Independent Python cross-check | Run guide | Local OHLCV fixture

01 / BUILD SUMMARY

A small, testable indicator toolkit

This development-stage package follows the supplied 23-page assignment PDF. Each required function keeps the exact name and argument signature expected by the evaluator. The implementations use only standard R operations: loops, arithmetic, indexing, vector allocation, and named lists.

The design keeps indicator code separate and readable, then adds a compact data utility layer for the stock-data, filtering, and trading-rule requirements. The test runner exercises sample calculations, output shapes, required error messages, two data frames, a date filter, and trading-rule labels.

Verification outcome: the native R 4.2.3 runner passed all nine indicators, required errors, OHLCV frames, time-frame/data-source filters, indicator stock selection, and the trading rule. The independent Python cross-check also passed.

Implementation map

Indicator	Core calculation	Output contract
SMA	Complete rolling windows; divide each sum by period.	Numeric vector, n - period + 1 values.
EMA	First value is data[1]; later values use multiplier 2/(period+1).	Numeric vector, same length as input.
MACD	Short EMA - long EMA; EMA signal; line - signal histogram.	Named list: macd_line, signal_line, histogram.
stdev	Population standard deviation with divisor n.	Single numeric value.
linreg	Assignment-defined subset, slope, intercept, fitted values.	Named list: slope, intercept, predicted_values.
RSI	Gain/loss vectors with Wilder smoothing.	Same-length vector with leading NA values.
StochRSI	Normalize valid RSI, then SMA for K and D.	Named list: k_line, d_line.
crossover	Detect Up or Down relationship changes.	String vector: Up, Down, None.
crossunder	Detect transition from >= to <.	String vector: None, True, False.

Direct-review files: R/sma.R, R/ema.R, R/macd.R, R/stdev.R, R/linreg.R, R/rsi.R, R/stoch_rsi.R, R/crossover.R, and R/crossunder.R.

02 / ALGORITHM NOTES

Moving averages and dispersion

SMA. The function allocates one output slot for every complete window. It checks the required short-input error before indexing.

EMA. The first input is the seed, matching the source pseudocode rather than a separately initialized simple average. The recursion is deterministic from the first row.

MACD. The script calls the required EMA calculation twice, subtracts the vectors element by element, then applies EMA again to the resulting line. It is the simplified version described in the PDF.

stdev. The implementation calculates the population value: the sum of squared deviations divided by n. It does not switch to the sample n - 1 convention.

Regression and momentum

linreg. The index window follows the supplied start and end formulas, including offset semantics. The return list exposes slope, intercept, and fitted values without adding statistical tests or error estimates.

RSI. Positive differences populate gains; non-positive differences populate absolute losses. Initial averages use the first period differences, then Wilder smoothing advances one observation at a time. Leading values remain NA.

StochRSI. Valid RSI observations are normalized to 0-1. Per the PDF, %K is the StochRSI and %D is the 3-day simple moving average of %K; this implementation preserves the required d_period argument. Undefined leading positions are retained, then SMA creates the shorter K and D lines.

Signals

crossover. The source pseudocode recognizes both upward and downward relationship changes, so the result uses Up, Down, and None strings. The first position is always None.

crossunder. The source pseudocode output is string-valued even though its prose also uses Boolean language. This implementation follows the exact template behavior: first None, then True or False.

Signature examples

```
sma(data, period) | ema(data, period) | macd(data, short_period, long_period, signal_period)
linreg(regressionSource, regressionLength, regressionOffset) | stoch_rsi(data, period, k_period, d_period)
```

03 / DATA AND RULES

Reproducible data frames

The local CSV contains 39 frozen OHLCV rows for two symbols: AAPL and MSFT. It is a clearly labeled reproducible fallback assembled from public sample archives, not a claimed retrieval from a reputable financial-data provider. R/data_utils.R converts date to Date, sorts by symbol and date, labels the source, and returns a named list of data frames. The required schema is symbol, date, open, high, low, close, and volume.

Fixture check	Verified value
Symbols	AAPL and MSFT
Rows	39 total
Required fields	date, open, high, low, close, volume
Data-source label	local_reproducible_fallback
Local reproducibility	No live network request is required by the tests

Source record

The cached rows and their public archive locations are documented in data/README.md. On 2026-09-01, quantmod 0.4.25 on R 4.2.3 attempted Yahoo Finance retrieval for AAPL and MSFT, but both requests failed because query2.finance.yahoo.com could not be resolved. No provider response was received, so no provider data is claimed or fabricated.

Filters and trading rule

filter_market_data accepts optional symbol, start-date, end-date, data-source, and daily/weekly/monthly time-frame filters. Weekly output uses Monday buckets; monthly output uses the first calendar day. OHLCV aggregation uses first open, maximum high, minimum low, last close, and summed volume. filter_stocks_by_indicator selects symbols using SMA, EMA, RSI, or stdev criteria. add_trading_rule aligns short and long SMAs to the original row count, then assigns Buy, Sell, or Hold.

This rule is deliberately transparent and reproducible. It demonstrates how computed indicators can be applied to filtered data without importing a technical-analysis package.

Example workflow

```
frames <- load_stock_data("data/stock_data.csv")
filtered <- filter_market_data(frames$MSFT, "2003-08-18", "2003-08-29", time_frame = "weekly", data_source =
"local_reproducible_fallback")
matches <- filter_stocks_by_indicator(frames, indicator = "sma", period = 3, operator = ">", threshold = 20)
signals <- add_trading_rule(filtered, short_period = 3, long_period = 5)
```

04 / TESTING AND RESULTS

Deterministic test coverage

The primary test file is tests/test_indicators.R. It uses stopifnot and tryCatch only, so it needs no testing package. It sources every indicator script, checks representative numeric values, validates named outputs and vector lengths, checks each required error message verbatim, and tests the expanded market-data utilities.

The independent fallback tests/crosscheck.py implements the equations separately in dependency-free Python. It also audits the R files for the nine exact function signatures and confirms both symbols and all required OHLCV columns in the fixture.

Verification layer	Status	Evidence
Python equation cross-check	PASS	9 indicator equations and sample outputs.
R signature audit	PASS	Exact names and argument lists found.
Required errors	PASS	SMA, linreg, crossover, crossunder messages.
Data / filters / rule	PASS	39 rows, 2 symbols, time frames, source filter, indicator filter, Buy/Sell/Hold.
Native R execution	PASS - R 4.2.3	Exact output saved in results/native_r_test_output.txt.

Representative numeric results

Example	Expected result
SMA, sample vector, period 3	12.333333333, 15.666666667, 17.666666667, 20, 21.666666667, 23.666666667, 23.333333333
EMA, sample vector, period 3	10, 11, 13, 16.5, 17.25, 19.625, 22.3125, 23.15625, 22.078125
stdev, sample vector	4.946628731 population standard deviation
linreg, c(1,3,2,5,4,6), 5, 1	slope 0.6; intercept 2; fitted 2.6, 3.2, 3.8, 4.4
RSI, supplied 10-value example, period 5	5 leading NA; then 50.526315789, 68.925619835, 71.835205993, 78.210793191, 66.859664061
StochRSI, supplied PDF call	data <- c(45, 50, 48, 55, 52, 49, 58, 60, 65, 62); stoch_rsi(data, period = 14, k_period = 3, d_period = 3). The PDF example is undersized for RSI period 14 and is documented as such.
Crossover / crossunder supplied arrays	Up at position 4; crossunder starts None and remains False

05 / RUN GUIDE

Run from the assignment directory

The directory is the folder containing R, data, tests, and submission.

Primary R run

```
R_LIBS_USER=/Users/sidhant/Desktop/pitte/users/sunny_khurana/BDA400/r-lib Rscript tests/test_indicators.R
```

Current final line: PASS: 9 indicators, required errors, OHLCV frames, time-frame/data-source filters, indicator stock filter, and trading rule.

Independent fallback when R is unavailable

```
python3 tests/crosscheck.py
```

Expected JSON status: PASS, with 9 checks, 39 dataset rows, and symbols AAPL and MSFT. This fallback is an independent audit, not a replacement for the final native R run on the submission machine.

Review checklist

Step	Action
1	Open README.md and this PDF for the package map and assumptions.
2	Run Rscript tests/test_indicators.R in a native R environment.
3	Compare the R output with results/indicator_results.md if needed.
4	Inspect data/stock_data.csv and R/data_utils.R for data-frame, filter, and rule coverage.
5	Replace the repository handoff token on the cover page only after a real repository URL exists.

Scope boundaries

The implementation deliberately avoids external R libraries for indicator calculations, professional-library MACD variations, statistical regression tests, and live network calls during tests. quantmod was used only for the documented provider-access attempt; the actual indicator code remains base R.

06 / REFERENCES AND HANDOFF

Assignment source

BDA400 / Data Science Tools and Techniques. *Assignment 5: Technical Analysis using R, Development Phase.* 23-page instruction PDF supplied in the local source directory. The implementation follows pages 2-20 for algorithms and submission expectations and pages 22-23 for the rubric.

Data archive record

AAPL public sample archive: https://github.com/mohabmes/pystocklib/blob/master/data/AAPL.csv
MSFT public sample archive: https://github.com/matplotlib/sample_data/blob/master/msft.csv

These public archive locations are recorded only to distinguish the cached fixture's origin. The fixture is not a live market feed or a reputable-provider export. If required for final submission, the student must replace it with a current provider export and record provider, symbols, date range, retrieval date, and export URL.

Final handoff

The local package is complete for review: code, tests, data, results, README, and this explanation PDF are under the assignment directory. Student action remains for the real repository URL, and provider-backed data replacement remains conditional on the instructor's submission requirement.

End of document